

Cloud Computing Lab

ASSIGNMENT - 5

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Batch : A3

Problem Statement:

Create a new docker image such as Apache Tomcat, Python, Flask, node.js without using any of the existing Docker Images.

Objectives:

1. To enable easy sharing of Images via Docker Hub
2. Run the same container on any machine where Docker is installed.
3. Simplifying dependency management without installation as configurations and libraries are pre-packaged inside Docker Images.
4. Build once; run anywhere, anytime.

```
ccew@fedora:~$ sudo dnf -y install dnf-plugins-core
```

```
[sudo] password for ccew:
```

```
Updating and loading repositories:
```

```
RPM Fusion for Fedora 42 - Nonfre 100% | 1.7 KiB/s | 15.6 KiB | 00m09s
```

```
RPM Fusion for Fedora 42 - Nonfre 100% | 2.2 KiB/s | 16.3 KiB | 00m07s
```

```
Fedora 42 - x86_64 - Updates 100% | 1.7 KiB/s | 10.7 KiB | 00m06s
```

```
Docker CE Stable - x86_64 100% | 1.5 KiB/s | 2.0 KiB | 00m01s
```

```
Copr repo for PyCharm owned by ph 100% | 1.6 KiB/s | 2.1 KiB | 00m01s
```

```
google-chrome 100% | 1.0 KiB/s | 1.3 KiB | 00m01s
```

```
Repositories loaded.
```

```
Package "dnf-plugins-core-4.10.1-1.fc42.noarch" is already installed.
```

```
Nothing to do.
```

```
ccew@fedora:~$ sudo dnf config-manager addrepo
```

```
--from-repofile=https://download.docker.com/linux/fedora/docker-ce.repo
```

```
File "/etc/yum.repos.d/docker-ce.repo" already exists and configures repositories with IDs
```

```
"docker-ce-stable docker-ce-stable-source docker-ce-test docker-ce-test-source". Add
```

```
"--overwrite" to overwrite.
```

```
ccew@fedora:~$ sudo dnf install docker-ce docker-ce-cli containerd.io
```

```
docker-buildx-plugin docker-compose-plugin -y
```

```
Updating and loading repositories:
```

```
Repositories loaded.
```

```
Package "docker-ce-3:29.2.1-1.fc42.x86_64" is already installed.
```

Package "docker-ce-cli-1:29.2.1-1.fc42.x86_64" is already installed.
Package "containerd.io-2.2.1-1.fc42.x86_64" is already installed.
Package "docker-buildx-plugin-0.31.1-1.fc42.x86_64" is already installed.
Package "docker-compose-plugin-5.0.2-1.fc42.x86_64" is already installed.

Nothing to do.

```
ccoew@fedora:~$ sudo systemctl start docker  
ccoew@fedora:~$ sudo systemctl enable docker
```

```
ccoew@fedora:~$ sudo usermod -aG docker $USER  
ccoew@fedora:~$ docker run hello-world
```

Hello from Docker!

This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:

```
$ docker run -it ubuntu bash
```

Share images, automate workflows, and more with a free Docker ID:

<https://hub.docker.com/>

For more examples and ideas, visit:

<https://docs.docker.com/get-started/>

```
ccoew@fedora:~$ mkdir docker-flask-app  
ccoew@fedora:~$ cd docker-flask-app
```

```
ccoew@fedora:~/docker-flask-app$ nano app.py  
app.py
```

```
from flask import Flask  
app = Flask(__name__)
```

```
@app.route('/')  
def home():  
    return "Hello Ma'am, this is my Docker Assignment!"
```

```
if __name__ == '__main__':  
    app.run(host='0.0.0.0', port=5000)
```

```
ccoew@fedora:~/docker-flask-app$ nano requirements.txt  
requirements.txt  
Flask
```

```
ccoew@fedora:~/docker-flask-app$ nano Dockerfile  
Docker file
```

```
ccoew@fedora:~/docker-flask-app$ docker login
```

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'

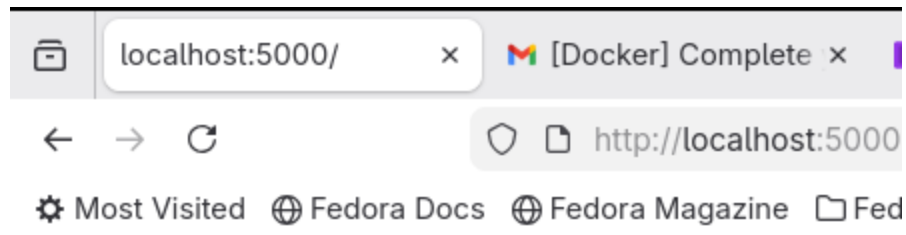
Your one-time device confirmation code is: CBCH-LCBT
Press ENTER to open your browser or submit your device code here:
<https://login.docker.com/activate>

Waiting for authentication in the browser...

WARNING! Your credentials are stored unencrypted in '/home/ccoew/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded

```
ccoew@fedora:~/docker-flask-app$ docker run -d -p 5000:5000 rutuja2811/flask-app  
e84e68a46fc0e41a28f02f5c3e164723b60941163002758274c0fa78a4e28388
```



```

ccoew@fedora:~/docker-flask-app$ docker build -t rutuja2811/flask-app .
[+] Building 9.1s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.4s
=> => transferring dockerfile: 205B                             0.0s
=> [internal] load metadata for docker.io/library/python:3.10-slim 2.3s
=> [auth] library/python:pull token for registry-1.docker.io    0.0s
=> [internal] load .dockerignore                                 0.4s
=> => transferring context: 2B                                   0.0s
=> [1/4] FROM
docker.io/library/python:3.10-slim@sha256:0ea978b26093c4f8cccbfc71da666d35e 0.7s
=> => resolve
docker.io/library/python:3.10-slim@sha256:0ea978b26093c4f8cccbfc71da666d35e 0.6s
=> [internal] load build context                                0.4s
=> => transferring context: 269B                                0.0s
=> CACHED [2/4] WORKDIR /app                                  0.0s
=> CACHED [3/4] COPY . .                                       0.0s
=> CACHED [4/4] RUN pip install -r requirements.txt            0.0s
=> exporting to image                                          2.0s
=> => exporting layers                                          0.0s
=> => exporting manifest
sha256:bff78c6c5d57451f6cb1c2afa9776808b80ab13615c5205844de7600f 0.1s
=> => exporting config
sha256:79075d928d61c00cf2566884d882b8e5853b11aee977c71088cf1f1cd65 0.1s
=> => exporting attestation manifest
sha256:19081d6577686a222516cc3c8da8e60a3cfedd6f2bbdc 0.8s
=> => exporting manifest list
sha256:dec1ac273c838cd03671aa1873e46894c11c9f2ee26d21f0f698 0.3s
=> => naming to docker.io/rutuja2811/flask-app:latest        0.1s
=> => unpacking to docker.io/rutuja2811/flask-app:latest     0.1s

```

```

ccoew@fedora:~/docker-flask-app$ docker push rutuja2811/flask-app

```

Using default tag: latest

The push refers to repository [docker.io/rutuja2811/flask-app]

451410c48d65: Pushed

fbd8fcca3db6: Pushed

bed04db721b4: Pushed

d6849cc06d47: Pushed

7dcb079764de: Pushed

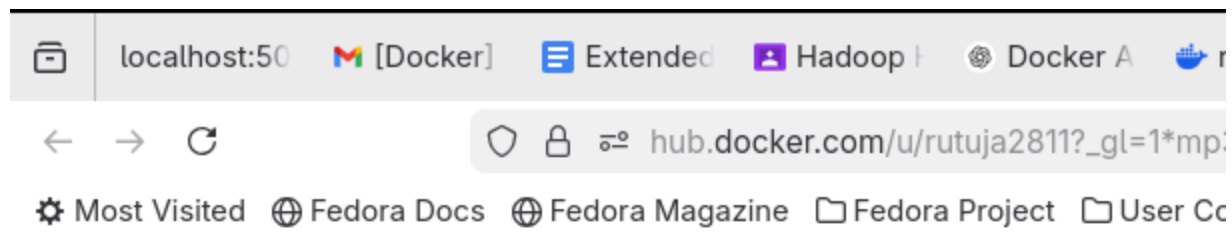
ddf3b231f567: Pushed

5435b2dcdf5c: Pushed

935d60c8480a: Pushed

latest: digest:

sha256:dec1ac273c838cd03671aa1873e46894c11c9f2ee26d21f0f698abf44a86c834 size: 856



rutuja2811 [Edit profile](#)

Community User

Repositories

Starred

Search by repository name

Displaying 1 to 1 of 1 repositories

IMAGE



rutuja2811/flask-app

rutuja2811

1m 0



Explore

Account

Resources

```
ccew@fedora:~/docker-flask-app$ docker logout
```

Removing login credentials for https://index.docker.io/v1/